

Nicholas Lee

Senior Agentic AI Engineer

Osseo, WI | +1 (713) 597-6928 | to.nicholasly@gmail.com | <https://www.linkedin.com/in/nick-lee-71b258430>

SUMMARY

Full stack AI engineer with 8+ years building production systems across React, Next.js, Node.js, and AWS, and deep hands-on experience shipping LLM-powered features and agentic workflows. Owns delivery from UI to backend to deployment, with a product mindset focused on scalable architecture, AI quality, and measurable business outcomes.

EXPERIENCE

Senior Agentic AI Engineer

Fermat · San Francisco, CA · Remote · Aug 2025 - Aug 2026

- Built an agentic content generation system for the AI Search Commerce Engine, using CrewAI to orchestrate multi-agent planning, tool selection, and execution for shoppable content, integrated with React, Next.js, Node.js, and PostgreSQL, reducing manual content production by 30%.
- Engineered autonomous refresh flows that detect declining AI-search performance using scheduled agents, LangChain for task decomposition, and OpenRouter for model routing, triggering regeneration pipelines with human-in-the-loop approval gates.
- Designed a multi-agent copilot (Pierre) that analyzes commerce data, surfaces insights, and initiates tasks, implemented with function calling, state management, and retry logic, deployed on AWS Lambda with observability via CloudWatch and PII-safe logging.
- Implemented RAG pipelines with vector embeddings and PostgreSQL pgvector for grounding product content, improving answer accuracy and citation quality, and integrated Claude and GPT-4 models through OpenRouter for cost/latency optimization.
- Created guardrails and validation layers to prevent hallucinated or off-brand content, including prompt injection defenses, output schema validation, and evaluation harnesses that track tool-call correctness and agent loop detection.
- Optimized production AI latency by caching embeddings, pre-warming model connections, and parallelizing independent agent steps, cutting average response time by 40% while maintaining 99.9% uptime.
- Built observability dashboards tracing each agent's steps, tool calls, and failures, enabling rapid debugging of multi-step workflows and reducing mean time to resolution by 50%.
- Collaborated with product managers and SMEs to refine requirements for AI features, document architecture decisions, and enforce best practices for AI-assisted coding, including code reviews and evaluation reviews.
- Owned end-to-end delivery of AI features from concept to production, balancing speed and maintainability while ensuring scalable architecture that handles growth from pilot to enterprise customers.

AI Software Engineer

Capital One · New York, NY · Hybrid · Jan 2023 - Apr 2025

- Built Chat Concierge, a proprietary multi-agent conversational AI assistant using Meta Llama, orchestrated with LangChain and tool calling to help car buyers compare vehicles and schedule test drives through natural language, deployed on AWS with React and Node.js frontends.
- Developed REST APIs for the Digital Wallet Service and Lounge Digital Pass using Node.js and NestJS, enabling contactless Apple/Google Wallet passes, with PostgreSQL persistence and AWS API Gateway for secure access.
- Designed a partner-agnostic point-of-sale API for Shop with Rewards, handling check, redeem, and refund flows across online and in-store channels, built with Java and .NET microservices and integrated with React dashboards.
- Implemented RAG-style retrieval over vehicle inventory and rewards data, using embeddings and vector search to ground responses, improving answer relevance and reducing hallucination rates.
- Created evaluation harnesses for conversational AI, testing multi-turn flows, tool-call accuracy, and context retention, and used results to refine prompts and model selection.
- Improved system reliability by adding circuit breakers and retry logic for LLM calls, reducing errors by 25%, and optimized token usage and latency to cut costs by 20%.
- Collaborated with product owners and data scientists to align AI features with business goals, document technical designs, and ensure compliance with financial regulations.
- Owned the full development lifecycle for AI features, from prototyping in Python to production deployment on AWS, with CI/CD pipelines and observability via Datadog.

Full Stack Engineer

DataChat · Madison, WI · On-site · May 2020 - Dec 2022

- Developed the DataChat Analytics Platform, a web-based tool for data exploration and visualization, using React, Next.js, and Node.js for the frontend and a backend running on AWS with PostgreSQL.
- Built DataChat Recipes/Workflows, a system for creating and reusing multi-step data transformations, implementing a backend service with Node.js and REST APIs, and a React UI for workflow management.
- Created collaborative sessions for organizing analyses and sharing results, using WebSockets for real-time updates and implementing role-based access control to ensure data privacy.
- Integrated PostgreSQL for data storage and retrieval, optimizing queries and indexing to support large datasets, and used Docker for containerization.
- Improved platform performance by profiling and optimizing frontend rendering and backend API response times, and implemented error handling and logging for production stability.
- Collaborated with data scientists to implement analytics features, documented API contracts, and owned feature delivery in agile sprints, including code reviews and deployment.

Software Engineer

Epic Systems · Verona, WI · On-site · Jul 2017 - Mar 2020

- Developed features for Epic Sonnet, a streamlined EHR, using C# and .NET for backend services, SQL Server for data, and JavaScript for frontend components, ensuring performance and reliability.
- Built Share Everywhere, a secure web application for patient data sharing, implementing REST APIs and integration with third-party systems using Node.js and PostgreSQL.
- Enhanced MyChart and mobile apps, adding features like scheduling and messaging, built with React and integrated with backend services on AWS.
- Collaborated with clinical stakeholders to refine requirements, wrote technical documentation, and participated in code reviews to maintain quality.
- Resolved production issues and optimized database queries, improving application response times and supporting 24/7 healthcare operations.

SKILLS

Programming Languages: TypeScript, JavaScript, Python, Java, C#, .NET, SQL

Frameworks & Libraries: React, Next.js, Node.js, NestJS, Express, Redux

AI & LLMs: LLM Integration, Prompt Engineering, Model Selection, Multi-model Orchestration, Claude, OpenRouter, LangChain, Meta Llama, CrewAI, Hugging Face, n8n, RAG Pipelines, Vector Databases, pgvector, Embeddings

Agents & Orchestration: Agentic AI, Multi-Agent Systems, Tool Calling, Function Calling, Agent Runtime, Workflow Automation, State Management, Memory, Orchestration, Human-in-the-Loop

Cloud & DevOps: AWS, Lambda, API Gateway, CloudFormation, Docker, CI/CD, Observability, CloudWatch, Datadog, Terraform

Databases & Data: PostgreSQL, MySQL, MongoDB, Redis, Data Modeling, ETL, SQL Optimization

Collaboration & Delivery: Agile, Scrum, Kanban, Code Reviews, Technical Documentation, Stakeholder Communication, Product Mindset, Ownership, Mentoring, AI Governance

EDUCATION

University of Wisconsin-Madison - Bachelor of Science in Computer Science, Aug 2013 ~ May 2017, Madison, WI, GPA: 3.95

CERTIFICATIONS

AWS Certified Cloud Practitioner | May 2024 - May 2027

AWS Certified Solutions Architect | Aug 2023 - Aug 2026

Microsoft Certified: Azure Fundamentals | May 2026

Crash Course on Python | Jul 2025